

Chapter 15: Vectors in Space (Vector trong Không gian)

Chapter Overview

Vectors are fundamental mathematical objects that have both **magnitude** (size) and **direction**. Unlike scalars, which only have magnitude (such as temperature or mass), vectors are used to represent quantities like force, velocity, and displacement. In three-dimensional space, vectors become essential tools for describing positions, movements, and relationships between objects.

In this chapter, we will:

- Define vectors in three-dimensional space and their properties
- Learn vector operations: addition, subtraction, and scalar multiplication
- Understand the geometric interpretation of vector operations
- Apply the parallelogram rule and triangle rule
- Work with the dot product (scalar product) and its applications
- Solve real-world problems involving forces, velocities, and displacements

15.1 Bilingual Vocabulary (Từ vựng Song ngữ)

Core Concepts

English	IPA	Vietnamese	Example
Vector	<i>/'vektər/</i>	Vectơ	<i>AB</i> represents a vector from A to B
Scalar	<i>/'skeɪlər/</i>	Đại lượng vô hướng	Temperature is a scalar quantity
Magnitude	<i>/'mæɡnɪtjuːd/</i>	Độ lớn	The magnitude of <i>v</i> is $ v $

Direction	/də'rekʃən/	Hướng	Vectors have both magnitude and direction
Position vector	/pə'zɪʃən 'vektər/	Vectơ vị trí	OP is the position vector of point P
Displacement vector	/dɪs'pleɪsmənt 'vektər/	Vectơ dịch chuyển	AB represents displacement from A to B
Zero vector	/'zɪərəʊ 'vektər/	Vectơ không	0 has magnitude 0
Unit vector	/'ju:nɪt 'vektər/	Vectơ đơn vị	A vector with magnitude 1
Opposite vector	/'ɒpəzɪt 'vektər/	Vectơ đối	$-a$ is the opposite of a
Equal vectors	/'i:kwəl 'vektərz/	Vectơ bằng nhau	Vectors with same magnitude and direction

Vector Operations

English	IPA	Vietnamese	Example
Vector addition	/'vektər ə'dɪʃən/	Phép cộng vectơ	$a + b$
Vector subtraction	/'vektər səb 'trækʃən/	Phép trừ vectơ	$a - b$
Scalar multiplication	/'skeɪlər ,mʌltɪplɪ 'keɪʃən/	Phép nhân vô hướng	ka where k is a scalar
Dot product	/dɒt 'prɒdʌkt/	Tích vô hướng	$\vec{a} \cdot \vec{b} =$

Parallel vectors	/ˈpærəlel ˈvektərz/	Vectơ cùng phương	$a = kb$ for some scalar k
Collinear vectors	/kəˈliɪniər ˈvektərz/	Vectơ cùng phương	Vectors lying on the same line
Perpendicular vectors	/ˌpɜːrpənˈdɪkjələr ˈvektərz/	Vectơ vuông góc	$a \cdot b = 0$

Geometric Terms

English	IPA	Vietnamese	Example
Parallelogram rule	/ˌpærəˈleləgræm ruːl/	Quy tắc hình bình hành	$AB + AD = AC$ in parallelogram ABCD
Triangle rule	/ˈtraɪæŋgəl ruːl/	Quy tắc tam giác	$AB + BC = AC$
Parallelepiped rule	/ˌpærəleləˈpaɪpɪd ruːl/	Quy tắc hình hộp	For 3D vector addition
Midpoint	/ˈmɪdpɔɪnt/	Trung điểm	M is the midpoint of AB
Centroid	/ˈsentɹɔɪd/	Trọng tâm	G is the centroid of triangle ABC

15.2 Vectors in Three-Dimensional Space (Vector trong Không gian Ba chiều)

Definition of a Vector

Definition: Vector

A **vector** is a directed line segment characterized by:

- Magnitude** (or length): the distance between its initial and terminal points
- Direction**: the orientation from the initial point to the terminal point

We denote a vector from point A to point B as

AB

or

$\overrightarrow{AB}$

.

- Point A is called the **initial point** (điểm đầu)
- Point B is called the **terminal point** (điểm cuối)

Notation:

- Bold lowercase letters: **a**, **b**, **v**

- Arrows over letters:

a

,

b

,

v

- Letters with arrows:

$\overrightarrow{AB}$

Magnitude of a Vector

The **magnitude** (or **length**) of vector

AB

is the distance from A to B, denoted:

$$|AB| = AB$$

For a vector

a

, we write

$|a|$

or

$\|a\|$

.

Equal Vectors

Definition: Equal Vectors

Two vectors

a

and

b

are **equal** if and only if:

1. They have the same magnitude:

$$|a| = |b|$$

2. They have the same direction

We write:

$$a = b$$

Important: Equal vectors do not need to have the same initial point. They can be translated in space while remaining equal.

Example 15.1: Equal Vectors in a Cube

Consider a cube ABCD. A'B'C'D' with edge length 1. Which of the following statements are true?

(a)

$$AA' = BC$$

(b)

$$AA' = CC'$$

(c)

$$AA' = B'B$$

Solution:

(a) The edges AA' and BC are skew lines (not coplanar), so vectors

$$AA'$$

The edges AA' and BC are skew lines (not coplanar), so vectors

AA'

and

BC

are not parallel. Therefore,

$AA' \neq BC$

. **False**

(b) Since $ACC'A'$ is a parallelogram, $AA' \parallel CC'$ and $AA' = CC'$. The vectors

AA'

and

CC'

have the same length and the same direction. Therefore,

$AA' = CC'$

. **True**

(c) Vectors

AA'

and

$B'B$

have the same length but opposite directions. Therefore,

$AA' \neq B'B$

. **False**

Answer: Only statement (b) is true.

Special Vectors

1. Zero Vector (

0

)

- A vector with magnitude 0
- Has no specific direction
- For any point A:
 $AA = 0$

2. Unit Vector

- A vector with magnitude 1
- Used to indicate direction only

3. Opposite Vectors

- Vectors

a

and

b

are opposite if they have the same magnitude but opposite directions

- We write:

$$b = -a$$

- Note:

$$BA = -AB$$

15.3 Vector Addition (Phép Cộng Vector)

Triangle Rule

Triangle Rule for Vector Addition

For any three points A, B, C in space:

$$AB + BC = AC$$

This is also called the **head-to-tail rule**: place the tail of the second vector at the head of the first vector.

Geometric interpretation: If you travel from A to B, then from B to C, the net displacement is from A to C.

Parallelogram Rule

Parallelogram Rule

If ABCD is a parallelogram, then:

$$AB + AD = AC$$

The sum of two adjacent sides equals the diagonal.

Properties of Vector Addition

For any vectors

a

,

b

,

c

in space:

1. **Commutative property:**

$$a + b = b + a$$

2. **Associative property:**

$$(a + b) + c = a + (b + c)$$

3. **Identity property:**

$$a + 0 = 0 + a = a$$

4. **Inverse property:**

$$a + (-a) = 0$$

Example 15.2: Vector Addition in a Cube

In a cube ABCD. A'B'C'D' with edge length 1, calculate the magnitude of

$$BC + DD'$$

.

Solution:

Since ABCD is a square,

$$BC = AD$$

.

Therefore:

$$BC + DD' = AD + DD' = AD'$$

Triangle ADD' is a right triangle with:

- $AD = 1$
(edge of cube)
- $DD' = 1$
(edge of cube)

Using the Pythagorean theorem:

$$|AD'| = \sqrt{AD^2 + DD'^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Answer:

$$|BC + DD'| = \sqrt{2}$$

Example 15.3: Triangle Rule Application

For a tetrahedron ABCD, prove that:

$$AC + BD = AD + BC$$

Solution:

Using the triangle rule, we can write:

$$AC = AD + DC$$

Therefore:

$$AC + BD = (AD + DC) + BD$$

$$= AD + (DC + BD)$$

$$= AD + (BD + DC)$$

(commutative property)

$$= AD + BC$$

(triangle rule)

Answer:

$$AC + BD = AD + BC$$

✓

Parallelepiped Rule (Quy tắc Hình hộp)

Parallelepiped Rule

For a parallelepiped (rectangular box) ABCD. A'B'C'D':

$$AB + AD + AA' = AC'$$

The sum of three edges from one vertex equals the space diagonal from that vertex.

This is the three-dimensional extension of the parallelogram rule.

Example 15.4: Parallelepiped Rule

In a parallelepiped ABCD. A'B'C'D', prove that:

$$BC + DC + AA' = AC'$$

Solution:

Since ABCD is a parallelogram:

- $BC = AD$
- $DC = AB$

Therefore:

$$BC + DC + AA' = AD + AB + AA'$$

$$= AB + AD + AA'$$

(commutative property)

$$= AC'$$

(parallelepiped rule)

Answer:

$$BC + DC + AA' = AC'$$

✓

15.4 Vector Subtraction (Phép Trừ Vector)

Definition

Vector Subtraction

The **difference** of two vectors

a

and

b

is defined as:

$$a - b = a + (-b)$$

We add the opposite of

b

to

a

.

Important Formula

For any three points O, A, B in space:

$$OB - OA = AB$$

Proof:

$$OB - OA = OB + (-OA) = OB + AO = AO + OB = AB$$

Example 15.5: Vector Subtraction

In a pyramid S. ABCD with parallelogram base ABCD, let M and N be the midpoints of AB and CD respectively. Prove that:

(a)

AM

and

CN

are opposite vectors

(b)

$$SC - AM - AN = SA$$

Solution:

(a) Since ABCD is a parallelogram, $AB \parallel CD$ and $AB = CD$.

Since M and N are midpoints:

$$\cdot AM = \frac{1}{2}AB$$

$$\cdot CN = \frac{1}{2}CD$$

Therefore

$$AM = CN$$

and $AM \parallel CN$.

However,

AM

and

CN

point in opposite directions (one points right, one points left).

Therefore:

$$CN = -AM$$

✓

(b) From part (a), we have

$$CN = -AM$$

, so

$$-AM = CN$$

.

$$SC - AM - AN = SC + CN - AN$$

$$= SN - AN$$

$$= SN + NA$$

$$= SA$$

✓

15.5 Scalar Multiplication (Phép Nhân Vô hướng)

Definition

Scalar Multiplication

For a real number

$$k \neq 0$$

and a vector

$$a \neq 0$$

, the product

$$ka$$

is a vector defined by:

1. Direction:

- Same direction as

$$a$$

if

$$k > 0$$

- Opposite direction to

$$a$$

if

$$k < 0$$

2. Magnitude:

$$|ka| = |k| \cdot |a|$$

Convention: If

$$k = 0$$

or

$$a = 0$$

, then

$$ka = 0$$

Collinear Vectors

Condition for Collinear Vectors

Two vectors

a

and

b

(with

$b \neq 0$)

are **collinear** (cùng phương) if and only if there exists a scalar

k

such that:

$$a = kb$$

Properties of Scalar Multiplication

For any scalars

h, k

and vectors

a, b

:

1. **Associative:**

$$h(ka) = (hk)a$$

2. **Distributive (scalars):**

$$(h + k)a = ha + ka$$

3. **Distributive (vectors):**

$$k(a + b) = ka + kb$$

4. **Identity:**

$$1 \cdot a = a$$

Example 15.6: Scalar Multiplication

In a triangular prism $ABC.A'B'C'$, let M and N be the midpoints of AB and AC respectively. Let O be the intersection of AB' and $A'B$. Prove that:

$$\vec{C'C} = (-2)\vec{OM}$$

Solution:

Since O is the midpoint of AB' , OM is the midsegment of triangle $AB'B$.

Therefore: $B'B \parallel OM$ and

$$\vec{B'B} = 2 \cdot \vec{OM}$$

Since $BCC'B'$ is a parallelogram: $B'B \parallel C'C$ and

$$\vec{B'B} = \vec{C'C}$$

Therefore: $C'C \parallel OM$ and

$$\vec{C'C} = 2 \cdot \vec{OM}$$

Since vectors

$$\vec{C'C}$$

and

$$\vec{OM}$$

have opposite directions:

$$\vec{C'C} = (-2)\vec{OM}$$

✓

15.6 The Dot Product (Tích Vô hướng)

Angle Between Two Vectors

Definition: Angle Between Vectors

The **angle between two non-zero vectors**

$\vec{a}$

and

b

is the angle

θ

formed when both vectors are drawn from the same initial point, where

$$0^\circ \leq \theta \leq 180^\circ$$

.

Notation:

(a, b)

or

$\angle(a, b)$

Special cases:

- If

a

and

b

have the same direction:

$$(a, b) = 0^\circ$$

- If

a

and

b

have opposite directions:

$$(a, b) = 180^\circ$$

- If

$a \perp b$

:

$$(a, b) = 90^\circ$$

Definition of Dot Product

Definition: Dot Product (Scalar Product)

For two non-zero vectors

a

and

b

, the **dot product** is a scalar defined by:

$$a \cdot b = |a| \cdot |b| \cdot \cos(a, b)$$

Convention: If

$$a = 0$$

or

$$b = 0$$

, then

$$a \cdot b = 0$$

Important Properties

1. Perpendicularity condition:

$$a \perp b \Leftrightarrow a \cdot b = 0$$

(for non-zero vectors)

2. Magnitude formula:

$$a \cdot a = |a|^2$$

3. Angle formula:

$$\cos(a, b) = \frac{a \cdot b}{|a| \cdot |b|}$$

Properties of the Dot Product

For any vectors

a, b, c

and scalar

k

:

1. Commutative:

$$a \cdot b = b \cdot a$$

2. **Scalar multiplication:**

$$k(a \cdot b) = (ka) \cdot b = a \cdot (kb)$$

3. **Distributive:**

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

Example 15.7: Dot Product in a Regular Tetrahedron

Consider a regular tetrahedron $S. ABCD$ where all edges have length

a

. Calculate:

(a)

$$AS \cdot BC$$

(b)

$$AS \cdot AC$$

Solution:

(a) Triangle SAD is equilateral with all sides equal to

a

, so

$$\angle SAD = 60^\circ$$

.

Since $ABCD$ is a square,

$$AD = BC$$

.

Therefore:

$$(AS, BC) = (AS, AD) = \angle SAD = 60^\circ$$

$$AS \cdot BC = |AS| \cdot |BC| \cdot \cos 60^\circ$$

$$= a \cdot a \cdot \frac{1}{2} = \frac{a^2}{2}$$

(b) Triangle SAC has

$$SA = SC = a$$

and

$$AC = a\sqrt{2}$$

(diagonal of square).

Using the law of cosines in triangle SAC:

$$AC^2 = SA^2 + SC^2 - 2 \cdot SA \cdot SC \cdot \cos(\angle SAC)$$

$$(a\sqrt{2})^2 = a^2 + a^2 - 2a^2 \cos(\angle SAC)$$

$$2a^2 = 2a^2 - 2a^2 \cos(\angle SAC)$$

$$\cos(\angle SAC) = 0$$

Therefore

$$\angle SAC = 90^\circ$$

, which means triangle SAC is a right isosceles triangle.

$$AS \cdot AC = |AS| \cdot |AC| \cdot \cos 45^\circ$$

$$= a \cdot a\sqrt{2} \cdot \frac{\sqrt{2}}{2} = a^2$$

Answer:

• (a)

$$AS \cdot BC = \frac{a^2}{2}$$

• (b)

$$AS \cdot AC = a^2$$

Example 15.8: Perpendicular Vectors

In a tetrahedron ABCD, AC and BD are both perpendicular to AB. Let M and N be the midpoints of AB and CD respectively. Prove that:

(a)

$$MN = \frac{1}{2}(AC + BD)$$

(b)

$$MN \cdot AB = 0$$

Solution:

(a) We have:

$$MN = MA + AC + CN$$

$$MN = MB + BD + DN$$

Adding these two equations:

$$2MN = (MA + MB) + (AC + BD) + (CN + DN)$$

Since M is the midpoint of AB:

$$MA + MB = 0$$

Since N is the midpoint of CD:

$$CN + DN = 0$$

Therefore:

$$2MN = AC + BD$$

$$MN = \frac{1}{2}(AC + BD)$$

✓

(b) From the given conditions:

$$AC \cdot AB = 0$$

and

$$BD \cdot AB = 0$$

Therefore:

$$MN \cdot AB = \frac{1}{2}(AC + BD) \cdot AB$$

$$= \frac{1}{2}(AC \cdot AB + BD \cdot AB)$$

$$= \frac{1}{2}(0 + 0) = 0$$

✓

15.7 Applications of Vectors (Ứng dụng của Vector)

Application 1: Forces in Equilibrium

Problem: Two forces

F_1

and

F_2

act on a point. Force

F_1

has magnitude 30 N and acts horizontally. Force

F_2

has magnitude 40 N and acts at an angle of 60° to

F_1

. Find:

(a) The magnitude of the resultant force

$$F = F_1 + F_2$$

(b) The angle between

F

and

F_1

Solution:

(a) Using the law of cosines:

$$|F|^2 = |F_1|^2 + |F_2|^2 + 2|F_1||F_2|\cos 60^\circ$$

$$= 30^2 + 40^2 + 2(30)(40) \cdot \frac{1}{2}$$

$$= 900 + 1600 + 1200 = 3700$$

$$|F| = \sqrt{3700} \approx 60.8 \text{ N}$$

(b) Using the law of sines in the triangle formed by

F_1

,

F_2

, and

F

:

$$\frac{|F_2|}{\sin \alpha} = \frac{|F|}{\sin 60^\circ}$$

where

α

is the angle between

F

and

F_1

.

$$\sin \alpha = \frac{|F_2| \sin 60^\circ}{|F|} = \frac{40 \cdot \frac{\sqrt{3}}{2}}{60.8} \approx 0.569$$

$$\alpha \approx 34.7^\circ$$

Answer:

- (a) The resultant force has magnitude approximately 60.8 N
- (b) The angle is approximately 34.7°

Application 2: Velocity and Displacement

Problem: A boat travels at 20 km/h in still water. The river current flows at 5 km/h perpendicular to the boat's direction. Find:

(a) The boat's actual velocity relative to the ground

(b) The angle at which the boat moves relative to its intended direction

Solution:

Let

v_b
= boat's velocity in still water = 20 km/h

Let

v_c
= current's velocity = 5 km/h (perpendicular)

(a) The actual velocity is:

$$v = v_b + v_c$$

Since the vectors are perpendicular:

$$|v| = \sqrt{|v_b|^2 + |v_c|^2} = \sqrt{20^2 + 5^2} = \sqrt{425} \approx 20.6 \text{ km/h}$$

(b) The angle

θ

satisfies:

$$\tan \theta = \frac{|v_c|}{|v_b|} = \frac{5}{20} = 0.25$$

$$\theta = \arctan(0.25) \approx 14.0^\circ$$

Answer:

- (a) The boat's actual velocity is approximately 20.6 km/h
- (b) The boat is pushed 14.0° off course by the current

15.8 Practice Exercises (Bài tập Luyện tập)

Exercise 15.1: In a parallelepiped ABCD. A'B'C'D', simplify:

- (a)
 $AB + BC + CC'$
- (b)
 $AB + AD + AA'$

Exercise 15.2: In a regular tetrahedron ABCD with edge length 2, calculate:

- (a)
 $AB \cdot AC$

- (b)
 $AB \cdot CD$

Exercise 15.3: Prove that in any tetrahedron ABCD:

$$AB + CD = AC + BD = AD + BC$$

Exercise 15.4: Two forces of magnitude 8 N and 6 N act at an angle of 90° to each other. Find the magnitude and direction of the resultant force.

15.9 Chapter Summary (Tóm tắt Chương)

Key Concepts

1. Vector Definition:

- A directed line segment with magnitude and direction
- Equal vectors: same magnitude and direction

2. Vector Operations:

- **Addition:** Triangle rule, parallelogram rule, parallelepiped rule
- **Subtraction:**

$$a - b = a + (-b)$$

- **Scalar multiplication:**

$$ka$$

changes magnitude by

$$|k|$$

3. Dot Product:

- $a \cdot b = |a| |b| \cos \theta$

- Perpendicularity:

$$a \perp b \Leftrightarrow a \cdot b = 0$$

Important Formulas

Formula	Description
$AB + BC = AC$	Triangle rule
$OB - OA = AB$	Vector subtraction
$a = kb$	Collinear vectors
$ \vec{a} \cdot \vec{b} \cos \theta = \vec{a} \cdot \vec{b}$	Scalar product
$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{ \vec{a} \vec{b} }$	Angle between vectors
$ \vec{a} ^2 = \vec{a} \cdot \vec{a}$	Magnitude of vector

Properties Summary

Vector Addition:

- Commutative:

$$a + b = b + a$$

- Associative:

$$(a + b) + c = a + (b + c)$$

Dot Product:

- Commutative:

$$a \cdot b = b \cdot a$$

- Distributive:

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

15.10 Review Exercises (Bài tập Ôn tập)

Review 15.1: In a cube ABCD. A'B'C'D' with edge length

a

:

- (a) Calculate
 $|AC + AA'|$
- (b) Calculate
 $AB \cdot AC$
- (c) Find the angle between
 AB
and
 AC

Review 15.2: In a regular triangular pyramid S. ABC where all edges equal

a

:

- (a) Calculate
 $SA \cdot SB$
- (b) Calculate
 $SA \cdot BC$
- (c) Prove that
 $SA + SB + SC$
is perpendicular to the base

Review 15.3: Prove that for any quadrilateral ABCD, if M, N, P, Q are the midpoints of AB, BC, CD, DA respectively, then MNPQ is a parallelogram.

Review 15.4: A plane flies at 500 km/h in still air. A wind blows from the west at 80 km/h. If the plane heads north, find:

- (a) The plane's actual velocity
 - (b) The direction the plane actually travels
-

Memory Aids (Gợi nhớ)

Vector operations mnemonic:

- **Addition:** "Head to tail" (triangle rule)
- **Parallelogram:** "Two sides give diagonal"
- **Parallelepiped:** "Three edges give space diagonal"

Dot product signs:

- $a \cdot b > 0$
→ acute angle ($< 90^\circ$)
- $a \cdot b = 0$
→ perpendicular ($= 90^\circ$)
- $a \cdot b < 0$
→ obtuse angle ($> 90^\circ$)

Common mistakes to avoid:

- Don't confuse
 AB
with
 BA
(they are opposite!)
- Equal vectors don't need the same starting point
- The dot product gives a **scalar**, not a vector
- When using
 $OB - OA = AB$
, the result points from A to B

End of Chapter 15